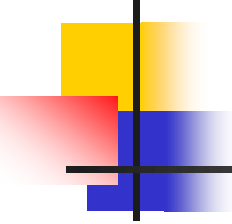




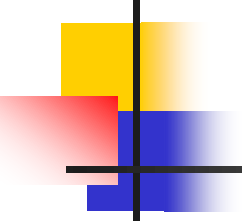
Java Evolution

- ❑ Java Introduction
- ❑ Java Features
- ❑ How Java Differs from other OO languages



Java - An Introduction

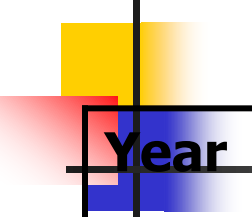
- Java - The new programming language developed by Sun Microsystems (James Gosling) in 1991.
- Originally called Oak.
- Java Authors: James Gosling, Arthur Van Hoff, Andy Bechtolsheim
- A general-purpose object-oriented language.
- Designed for easy Web/Internet applications



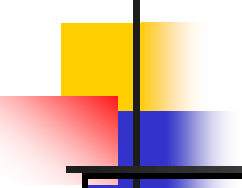
Java - An Introduction

- Originally created for consumer electronics (TV, VCR, Freeze, Washing Machine, Mobile Phone).
- Java - CPU Independent language
- Internet and Web was just emerging, so Sun turned it into a language of Internet Programming.
- It allows you to publish a webpage with Java code in it.

Java Milestones

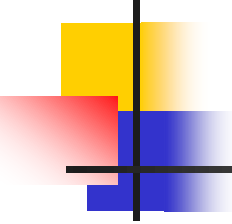


Year	Development
1990	Sun decided to developed special software that could be used for electronic devices. A project called Green Project created and head by James Gosling.
1991	Explored possibility of using C++, with some updates announced a new language named "Oak"
1992	The team demonstrated the application of their new language to control a list of home appliances using a hand held device.
1993	The World Wide Web appeared on the Internet and transformed the text-based interface to a graphical rich environment. The team developed Web applets (time programs) that could run on all types of computers connected to the Internet.



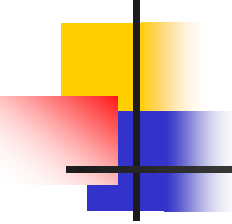
Java Milestones

Year	Development
1994	The team developed a new Web browser called "Hot Java" to locate and run Applets. HotJava gained instant success.
1995	Oak was renamed to Java, as it did not survive "legal" registration. Many companies such as Netscape and Microsoft announced their support for Java
1996	Java established itself as both 1. "the language for Internet programming" 2. a general purpose OO language.
1997	A class libraries, Community effort and standardization, Enterprise Java, Clustering, etc..



Java Features

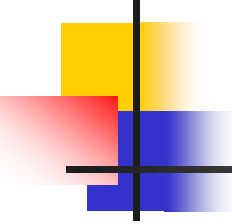
- Familiar, Simple, Small
- Compiled and Interpreted
- Platform-Independent and Portable
- Object-Oriented
- Robust and Secure
- Distributed
- Multithreaded
- High Performance
- Dynamic and Extensible



Cont...

- **Familiar, Simple & small**

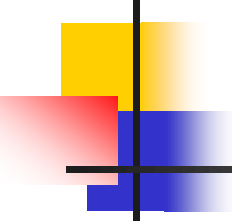
- Java does not use Preprocessor header files, goto statement and many others
- no pointers
- automatic garbage collection
- rich pre-defined class library
<http://java.sun.com/j2se/1.4.2/docs/api/>



Cont...

- Object-Oriented

- focus on the data (objects) and methods manipulating the data
- Almost every thing in Java is Object.
- all functions are associated with objects
- almost all datatypes are objects (files, strings, etc.)
- potentially better code organization and reuse



Cont...

- Platform-Independent and Portable

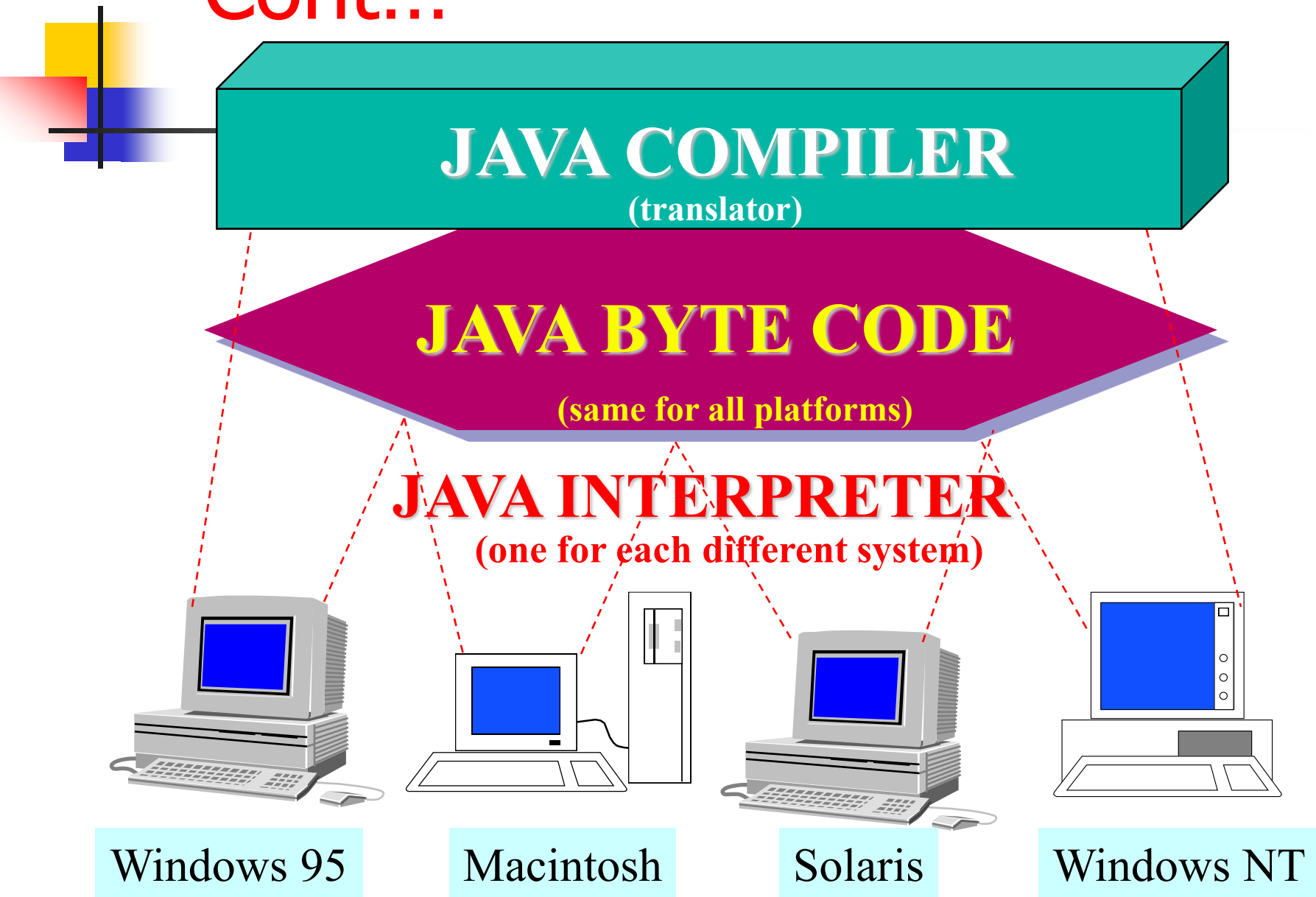
- Java is portable.

- 1) Java Compiler generates bytecode that can be implemented on any machine.

- 2) the sizes of the primitive data types are always the same

- application runs on all platforms

Cont...



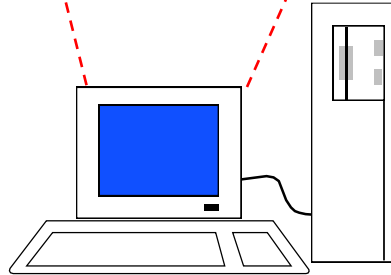
JAVA COMPILER
(translator)

JAVA BYTE CODE
(same for all platforms)

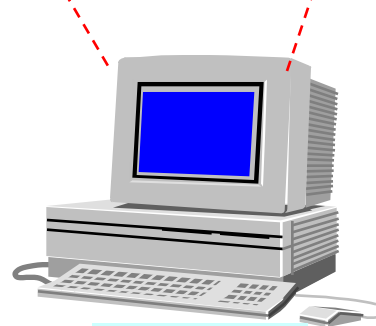
JAVA INTERPRETER
(one for each different system)



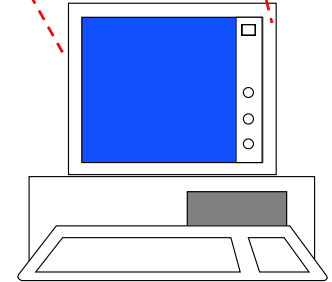
Windows 95



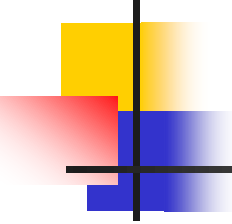
Macintosh



Solaris



Windows NT

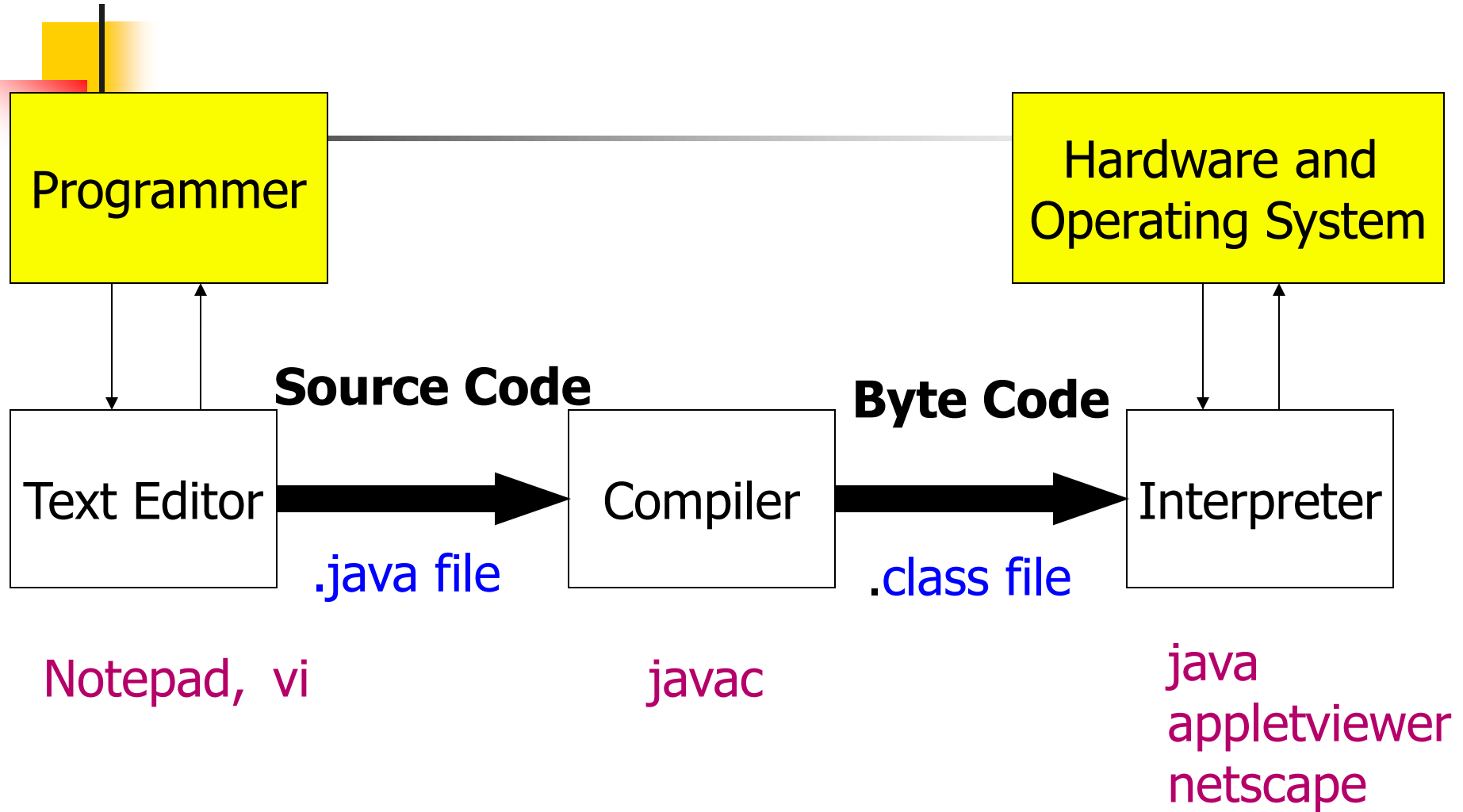


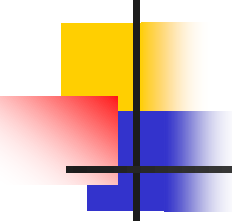
Cont...

■ Compiled and Interpreted

- java compiler generate byte-codes, not native machine code
- the compiled byte-codes are platform-independent
- java bytecodes are translated on the fly to machine readable instructions in runtime (Java Virtual Machine)

Cont...

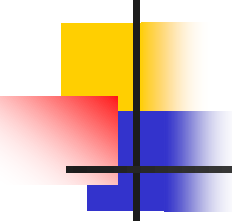




Cont...

- Robust and Secure

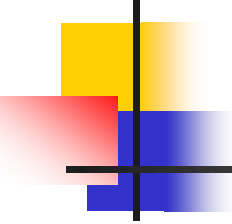
- usage in networked environments requires more security
- Provides many safeguard to ensure reliable code.
- Strict compile time & run time type checking.
- memory allocation model is a major defense
- access restrictions are forced (private, public)



Cont...

- Distributed

- Designed as distributed lang. For creating application on network.
- Java program can access remote access remote objects on internet

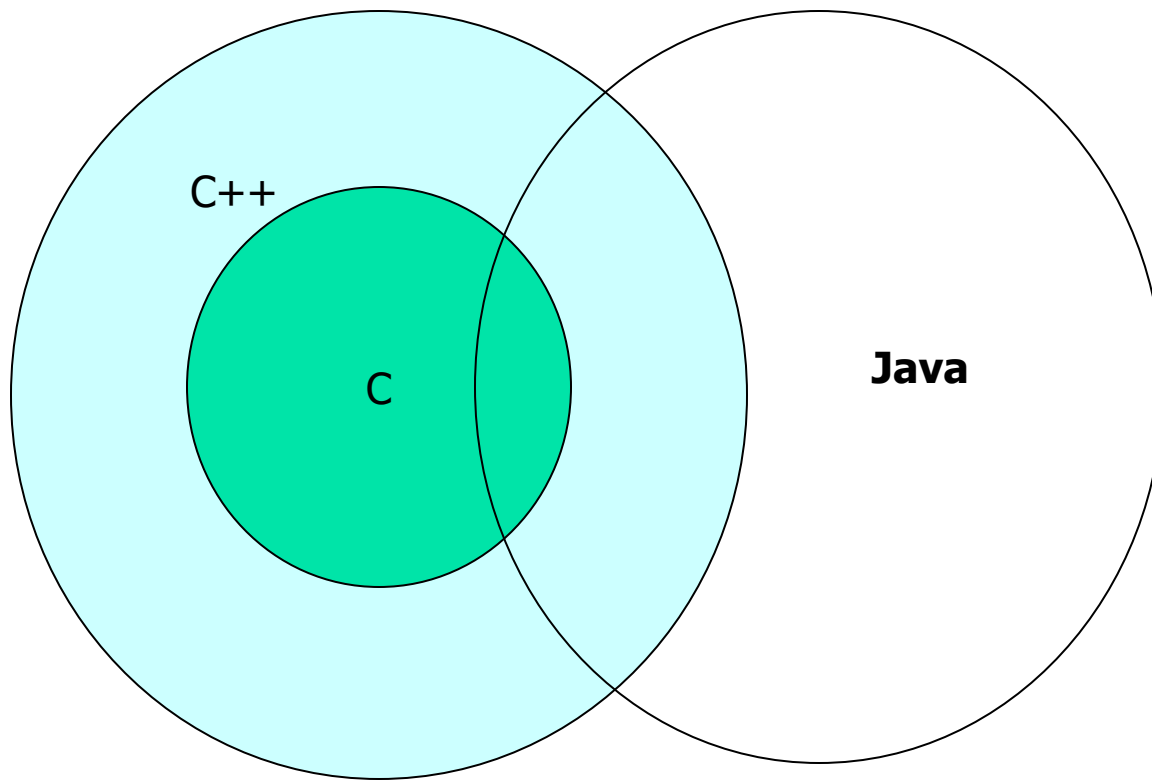


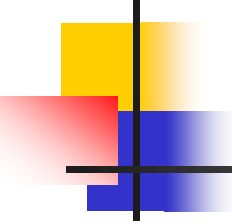
Cont...

■ Multithreaded

- Handles multiple task simultaneously
- Java tools support multiprocessing synchronization.

How Does Java Compare to C++ and Other OO Languages





Java better than C ?

- No **Preprocessor**
- No Global Variables
- No data types **struct, union and enum**
- No Goto ,size of and typedef
- No type modifier keywords
uto,extern,register,signed and unsigned.
- No Pointers
- Add new operator instance of and >>>
- Add labelled **break** and **continue**